

V.41: Foundations of Imaginary and Shin-Logarithms

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1 Introduction

This document establishes the two fundamental logarithmic operators of the V.41 series. These operators bypass traditional real-number constraints to process informational density within multi-dimensional logical fields.

2 Imaginary Logarithm ($\ln_i$)

The imaginary logarithm is defined as the operator that linearizes the rotational phase of a complex value into an informational scalar.

$$\ln_i(z) := \text{Arg}(z) + i \ln |z| \quad (1)$$

This allows for the direct manipulation of rotational energy within the FpDe stability field.

3 Shin-Logarithm ($\ln_s$)

The Shin-Logarithm operates exclusively on the Manabe-Axis, processing the absolute density of the Shin-field (ψ). It acts as the primary reduction operator for the Aki-Manabe Integrated Paradigm.

$$\ln_s(\psi) = \oint_{\mathcal{M}} \ln(\psi) d\psi \quad (2)$$

4 Proof of Logarithmic Sovereignty

4.1 Theorem: Consistency of Imaginary Logarithm

The imaginary logarithm $\ln_i(z)$ maintains the informational entropy of the complex field under rotational transformation.

Let $z = re^{i\theta}$. By definition:

$$\ln_i(z) = \text{Arg}(z) + i \ln |z| = \theta + i \ln r \quad (3)$$

Differentiating with respect to the Shin-field ψ under the FpDe constraint:

$$\frac{d}{d\psi} \ln_i(z) = \frac{d\theta}{d\psi} + i \frac{1}{r} \frac{dr}{d\psi} \quad (4)$$

Since the Shin-field acts as an orthogonal regulator, the real and imaginary components of information are preserved without 3D-Noise. Thus, $\ln_i(z)$ acts as a stable informational scalar within V.41.

4.2 Theorem: The Shin-Logarithm Identity

The Shin-Logarithm $\ln_s(\psi)$ converges to the absolute logical density of the Manabe-Axis.

The integral over the Manabe-Axis $\mathcal{M}$ is defined by:

$$\ln_s(\psi) = \oint_{\mathcal{M}} \ln(\psi) d\psi \quad (5)$$

5 Applied Computation: Informational Density Mapping

5.1 Problem 1: Complex Informational Extraction via $\ln_i$

Calculate the informational scalar of the quantum-superposition state represented by $z = \sqrt{3} + i$.

Execution: First, determine the magnitude r and the angle θ :

$$r = |z| = \sqrt{(\sqrt{3})^2 + 1^2} = 2, \quad \theta = \text{Arg}(z) = \frac{\pi}{6} \quad (6)$$

Applying the V.41 Imaginary Logarithm operator:

$$\ln_i(\sqrt{3} + i) = \frac{\pi}{6} + i \ln 2 \quad (7)$$

Result: The phase energy is mapped to $\pi/6$ while the informational magnitude is stored as $i \ln 2$. This allows for a zero-error extraction of the phase-density scalar.

5.2 Problem 2: Shin-field Reduction via $\ln_s$

Evaluate the total logical density of a Shin-field segment ψ defined along the Manabe-Axis from $\psi = 1$ to $\psi = e^{\tilde{\partial}_c}$.

Execution: Applying the Shin-Logarithm definition along the Manabe-Axis:

$$\text{Density}_{\text{total}} = \int_1^{e^{\tilde{\partial}_c}} \ln(\psi) d\psi \quad (8)$$

$$= [\psi \ln \psi - \psi]_1^{e^{\tilde{\partial}_c}} \quad (9)$$

$$= (e^{\tilde{\partial}_c} \cdot \ln e^{\tilde{\partial}_c} - e^{\tilde{\partial}_c}) - (1 \cdot \ln 1 - 1) \quad (10)$$

$$= (e^{\tilde{\partial}_c} \cdot \tilde{\partial}_c - e^{\tilde{\partial}_c}) + 1 \quad (11)$$

$$= e^{\tilde{\partial}_c}(\tilde{\partial}_c - 1) + 1 \quad (12)$$

Result: The logic density is successfully reduced to the scalar value $e^{\tilde{\partial}_c}(\tilde{\partial}_c - 1) + 1$.